Outputs

Question01:

```
Enter number of elements (at least 10): 10
Enter 10 elements: 8 4 7 2 9 1 3 6 8 10

--- QUICK SORT ---
Selected pivot: 10
Left subarray: [8, 4, 7, 2, 9, 1, 3, 6, 8]
Pivot: [10]
Right subarray: []

Selected pivot: 8
Left subarray: [8, 4, 7, 2, 1, 3, 6]
Pivot: [8]
Right subarray: [9]

Selected pivot: 6
Left subarray: [4, 2, 1, 3]
Pivot: [6]
Right subarray: [8, 7]

Selected pivot: 3
Left subarray: [2, 1]
Pivot: [3]
Right subarray: [4]

Selected pivot: 1
Left subarray: []
Pivot: [1]
Right subarray: [2]

Selected pivot: 7
Left subarray: []
Pivot: [7]
Right subarray: [8]

Total subarrays generated by Quick Sort: 9
Final sorted array: [1, 2, 3, 4, 6, 7, 8, 8, 9, 10]
```

```
--- MERGE SORT ---
Dividing [8, 4, 7, 2, 9, 1, 3, 6, 8, 10] into [8, 4, 7, 2, 9] and [1, 3, 6, 8, 10]
Dividing [8, 4, 7, 2, 9] into [8, 4, 7] and [2, 9]
Dividing [8, 4, 7] into [8, 4] and [7]
Dividing [8, 4] into [8] and [4]
Merged subarray: [4, 8]
Merged subarray: [4, 7, 8]
Dividing [2, 9] into [2] and [9]
Merged subarray: [2, 9]
Merged subarray: [2, 4, 7, 8, 9]
Dividing [1, 3, 6, 8, 10] into [1, 3, 6] and [8, 10]
Dividing [1, 3, 6] into [1, 3] and [6]
Dividing [1, 3] into [1] and [3]
Merged subarray: [1, 3]
Merged subarray: [1, 3, 6]
Dividing [8, 10] into [8] and [10]
Merged subarray: [8, 10]
Merged subarray: [1, 3, 6, 8, 10]
Merged subarray: [1, 2, 3, 4, 6, 7, 8, 8, 9, 10]
Total subarrays generated during Merge Sort division: 18
Final sorted array: [1, 2, 3, 4, 6, 7, 8, 8, 9, 10]


...Program finished with exit code 0
Press ENTER to exit console.
```

Question02:

```
Enter number of product IDs (at least 20): 21
Enter product IDs: 9 12 15 31 2 26 97 56 72 36 52 38 17 22 91 85 55 14 20 5 63

Level 1: Quick Sort Partitioning
Pivot: 63
Left subarray: [9, 12, 15, 31, 2, 26, 56, 36, 52, 38, 17, 22, 55, 14, 20, 5]
Pivot subarray: [63]
Right subarray: [97, 91, 85, 72]

Level 2: Merge Sort Division
Left subarray: [9, 12, 15, 31, 2, 26, 56, 36]
Right subarray: [52, 38, 17, 22, 55, 14, 20, 5]

Level 3: Quick Sort Partitioning
Pivot: 36
Left subarray: [9, 12, 15, 31, 2, 26]
Pivot subarray: [36]
Right subarray: [56]

Level 4: Merge Sort Division
Left subarray: [9, 12, 15]
Right subarray: [31, 2, 26]

Level 5: Quick Sort Partitioning
Pivot: 15
Left subarray: [9, 12]
Pivot subarray: [15]
Right subarray: []

Level 6: Merge Sort Division
Left subarray: [9]
Right subarray: [12]
Merged subarray: [9, 12]

Level 5: Quick Sort Partitioning
Pivot: 26
Left subarray: [2]
```

```
Level 5: Quick Sort Partitioning
Pivot: 26
Left subarray: [2]
Pivot subarray: [26]
Right subarray: [31]
Merged subarray: [2, 9, 12, 15, 26, 31]

Level 3: Quick Sort Partitioning
Pivot: 5
Left subarray: []
Pivot subarray: [5]
Right subarray: [38, 17, 22, 55, 14, 20, 52]

Level 4: Merge Sort Division
Left subarray: [38, 17, 22, 55]
Right subarray: [14, 20, 52]

Level 5: Quick Sort Partitioning
Pivot: 55
Left subarray: [38, 17, 22]
Pivot subarray: [55]
Right subarray: []

Level 6: Merge Sort Division
Left subarray: [38, 17]
Right subarray: [22]

Level 7: Quick Sort Partitioning
Pivot: 17
Left subarray: []
Pivot subarray: [17]
Right subarray: [38]
Merged subarray: [17, 22, 38]

Level 5: Quick Sort Partitioning
Pivot: 52
Left subarray: [14, 20]
```

```
Level 5: Quick Sort Partitioning
Pivot: 52
Left subarray: [14, 20]
Pivot subarray: [52]
Right subarray: []

Level 6: Merge Sort Division
Left subarray: [14]
Right subarray: [20]
Merged subarray: [14, 20]
Merged subarray: [14, 17, 20, 22, 38, 52, 55]
Merged subarray: [2, 5, 9, 12, 14, 15, 17, 20, 22, 26, 31, 36, 38, 52, 55, 56]

Level 2: Merge Sort Division
Left subarray: [97, 91]
Right subarray: [85, 72]

Level 3: Quick Sort Partitioning
Pivot: 91
Left subarray: []
Pivot subarray: [91]
Right subarray: [97]

Level 3: Quick Sort Partitioning
Pivot: 72
Left subarray: []
Pivot subarray: [72]
Right subarray: [85]
Merged subarray: [72, 85, 91, 97]

--- SUBARRAY COUNTS ---
Level 1: 3 subarrays generated
Level 2: 4 subarrays generated
Level 3: 9 subarrays generated
Level 4: 4 subarrays generated
Level 5: 9 subarrays generated
Level 6: 6 subarrays generated
```

```
Merged subarray: [72, 85, 91, 97]

--- SUBARRAY COUNTS ---
Level 1: 3 subarrays generated
Level 2: 4 subarrays generated
Level 3: 9 subarrays generated
Level 4: 4 subarrays generated
Level 5: 9 subarrays generated
Level 6: 6 subarrays generated
Level 7: 2 subarrays generated

Final sorted product IDs: [2, 5, 9, 12, 14, 15, 17, 20, 22, 26, 31, 36, 38, 52, 55, 56, 63, 72, 85, 91, 97]


...Program finished with exit code 0
Press ENTER to exit console.
```

Question03:

```
Enter number of students (at least 20): 20
Enter the last four digits of student IDs: 1234 4567 2347 6812 2387 5432 3214 6782 9123 4513 7864 6547 1209 2168 3257 2306 8543 7621 9072 8561

After units pass: 7621 8561 6812 5432 6782 9072 9123 4513 8543 1234 3214 7864 2306 4567 2347 2387 6547 3257 2168 1209
IDs with even digit at this position: 9

After tens pass: 2306 1209 6812 4513 3214 7621 9123 5432 1234 8543 2347 6547 3257 8561 7864 4567 2168 9072 6782 2387
IDs with even digit at this position: 13

After hundreds pass: 9072 9123 2168 1209 3214 1234 3257 2306 2347 2387 5432 4513 8543 6547 8561 4567 7621 6782 6812 7864
IDs with even digit at this position: 9

After thousands pass: 1209 1234 2168 2306 2347 2387 3214 3257 4513 4567 5432 6547 6782 6812 7621 7864 8543 8561 9072 9123
IDs with even digit at this position: 11

Sorted IDs: 1209 1234 2168 2306 2347 2387 3214 3257 4513 4567 5432 6547 6782 6812 7621 7864 8543 8561 9072 9123

Duplicate IDs and their frequencies:
No duplicate IDs found.

Digit sums:
1209 = 12
1234 = 10
2168 = 17
2306 = 11
2347 = 16
2387 = 20
3214 = 10
3257 = 17
4513 = 13
4567 = 22
5432 = 14
6547 = 22
6782 = 23
```

```
5432 = 14
6547 = 22
6782 = 23
6812 = 17
7621 = 16
7864 = 25
8543 = 20
8561 = 20
9072 = 18
9123 = 15
ID with highest digit sum: 7864 (sum = 25)

Longest sequence with even digit sums: 2347 2387 3214
Sequence length: 3


...Program finished with exit code 0
Press ENTER to exit console.
```

Question04:

```
Enter the number of elements (at least 10): 10
Enter decimal integers:
12 5 9 3 12 7 8 15 6 10

Binary Representations:
12 = 1100
5 = 0101
9 = 1001
3 = 0011
12 = 1100
7 = 0111
8 = 1000
15 = 1111
6 = 0110
10 = 1010

==================================
Pass 1 - Bit Position 0 (LSB)
==================================
Zeros: 5
Ones: 5
Decimal: 12 12 8 6 10 5 9 3 7 15
Binary:  1100 1100 1000 0110 1010 0101 1001 0011 0111 1111

==================================
Pass 2 - Bit Position 1
==================================
Zeros: 5
Ones: 5
Decimal: 12 12 8 5 9 6 10 3 7 15
Binary:  1100 1100 1000 0101 1001 0110 1010 0011 0111 1111

==================================
Pass 3 - Bit Position 2
==================================
Zeros: 4
Ones: 6
```

```
Zeros: 4
Ones: 6
Decimal: 8 9 10 3 12 12 5 6 7 15
Binary:  1000 1001 1010 0011 1100 1100 0101 0110 0111 1111

==================================
Pass 4 - Bit Position 3 (MSB)
==================================
Zeros: 4
Ones: 6
Decimal: 3 5 6 7 8 9 10 12 12 15
Binary:  0011 0101 0110 0111 1000 1001 1010 1100 1100 1111

==================================
Final Sorted Array
==================================
Decimal: 3 5 6 7 8 9 10 12 12 15
Binary:  0011 0101 0110 0111 1000 1001 1010 1100 1100 1111

==================================
Additional Analysis
==================================
Longest Binary Representation:
15 = 1111
Length = 4 bits

Largest Input Value:
15 = 1111
Number of set bits (1s): 4

Numbers with the same number of set bits:
15
Count: 1


...Program finished with exit code 0
Press ENTER to exit console.
```